

CS 736: Medical Image Computing

(Instructor : Suyash P. Awate)

Mid-Semester Examination (Maximum Points 65)

Date: 28 Feb 2019
Duration: 4 pm - 6 pm

Roll Number: _____

Name: _____

When in doubt, make suitable assumptions, state them clearly, and proceed to solve the problem.
All the best!

Formulae for Reference

Radon Transform: The Radon transform of a function $f(x, y)$ is a function $Rf(t, \theta)$ where for any given t, θ , the value of the Radon transform is

$$Rf(t, \theta) := \int_{s=-\infty}^{\infty} f(x(s), y(s)) ds, \text{ where } x(s) := t \cos \theta - s \sin \theta, y(s) := t \sin \theta + s \cos \theta$$

Poisson probability mass function: $P(x; \lambda) := \lambda^x \exp(-\lambda)/x!$ with mean = λ , variance = λ , and mode = $\text{floor}(\lambda)$.

Rician probability density function: $P(x; \nu, \sigma) := \frac{x}{\sigma^2} \exp(-\frac{x^2 + \nu^2}{2\sigma^2}) I_0(\frac{x\nu}{\sigma^2}), \forall x \geq 0$

Gamma probability mass function: $P(x; \alpha, \beta) := \eta(\alpha, \beta) x^{\alpha-1} \exp(-\beta x)$, where $\eta(\alpha, \beta)$ is a normalization constant that is a function of parameters α and β . In this case, mean = α/β , variance = α/β^2 , and mode = $(\alpha - 1)/\beta$.

Rayleigh probability density function: $P(x; \sigma) := \frac{x}{\sigma^2} \exp(-\frac{x^2}{2\sigma^2}), \forall x \geq 0$

1. [5 points] Describe the algorithm for simulated annealing to find the mode of a probability density function.

2. [20 points] Consider a color image (3-channel; RGB) acquired by a microscope. The specific biological specimen under the microscope is such that, at every spatial location in the specimen,
- (i) the photon intensity emanating from the specimen in the R and G frequency bands of the visible spectrum is such that the R-channel intensity is a fixed (but unknown) linear / affine function of the G-channel intensity, and
 - (ii) the photon intensity emanating in the B channel is independent of the photon intensity in the R and G channels.

At any spatial location $x \in \mathbb{R}^2$ within the biological specimen, consider the aforementioned linear / affine function to be of the form $R(x) := mG(x) + c$, where the parameters $m \in \mathbb{R}$ and $c \in \mathbb{R}$ are (i) unknown and (ii) independent of the spatial location x .

The noise introduced by the scanner can be approximated / assumed to be independent and identically distributed (i.i.d.) zero-mean Gaussian for each color channel.

- (5 points) Formulate an edge-preserving Markov random field (MRF) prior model for the uncorrupted image.

- (5 points) Formulate a likelihood model for the imaging process.

- (5 points) Give a **single** Bayesian formulation for inferring the uncorrupted RGB image directly from the corrupted data.

- (5 points) Clearly describe an implementable algorithm for optimization.

3. [20 points] Consider the task of fitting (i) a N -point pointset representation of shape $x := \{x_n \in \mathbb{R}^2\}_{n=1}^N$, via a similarity transform, to (ii) a set of N landmarks $y := \{y_n \in \mathbb{R}^2\}_{n=1}^N$ placed manually on an object boundary in the image.

The manual point placement is, however, erroneous and it is known that the errors are in the form of independent and identically distributed Gaussian in the coordinates.

Nevertheless, there is a statistical shape prior model learned on the shape pointsets, in the form of a mean shape μ and a covariance of shapes C . Furthermore, it is known that the variability in the shape of objects has exactly two modes of variation.

- (5 points) Design a mathematical representation of the pointset x in terms of the prior model parameters and using all the prior information. Your model must be such that the algorithm that you design for the task must never lead to numerical issues, e.g., values of variables in the code becoming infinity or NaN.

- (5 points) Design a statistical prior model $P(x)$.

- (5 points) Design a mathematical representation of the likelihood model.

- (5 points) Design an implementable algorithm for optimizing the model-fitting process using the prior model and the likelihood model. Your algorithm must never lead to numerical issues, e.g., values of variables in the code becoming infinity or NaN.

4. [8 points] Consider a X-ray beam passing through a composite material comprising three layers, from $x \rightarrow -\infty$ to $x \rightarrow +\infty$.

The attenuation coefficients in the material are given as:

- (i) $A(x) := 1 \text{ millimeter}^{-1}$, for $x \in [-1, 0] \text{ centimeter}$;
- (ii) $A(x) := (1 + x) \text{ centimeter}^{-1}$, for $x \in [0, 4] \text{ centimeter}$;
- (iii) $A(x) := (9 - x) \text{ centimeter}^{-1}$, for $x \in [4, 8] \text{ centimeter}$;
- (iv) $A(x) := 0$ elsewhere.

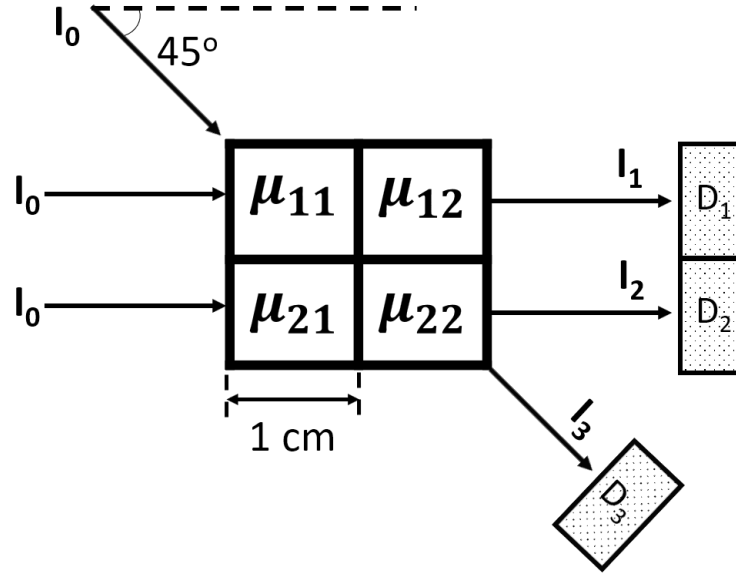
If the energy of the X-ray beam incoming from $x \rightarrow -\infty$ is I , what is the intensity of the X-ray beam at:

- (4 points) the spatial location $x = 0 \text{ cm}$.

- (2 points) the spatial location $x = 4$ cm.

- (2 points) the spatial location $x = 8$ cm.

5. [12 points] Consider a simple X-ray CT acquisition setup shown below. Three X-ray beams with initial energy I_0 are passed through the specimen, represented by a 2×2 discretized grid (with square pixels of size $1 \text{ cm} \times 1 \text{ cm}$), to produce the beam-energy measurements $\{I_1, I_2, I_3\}$ at detectors $\{D_1, D_2, D_3\}$ respectively. The measurements are corrupted such that the noise in $\log(I_1/I_0)$, $\log(I_2/I_0)$, and $\log(I_3/I_0)$ is assumed to be additive zero-mean independent Gaussian. CT reconstruction estimates the attenuation coefficients (ACs) $\{\mu_{11}, \mu_{12}, \mu_{21}, \mu_{22}\} \text{ cm}^{-1}$ from the data.



- (2 points) Design a prior $P_1(\cdot)$ on the 2×2 image of AC values based on the information that only a few of those AC values are non-zero.

- (2 points) Design a Markov random field prior $P_2(\cdot)$ on the 2×2 image of AC values, using a 4-neighborhood system and assuming that the image is smooth and devoid of any edges.

- (2 points) Design an optimization problem to solve for the AC values using both (i) the prior model $P_1(\cdot)$ and (ii) the data.

- (6 points: 2 + 4) Design an optimization problem to solve for the AC values using both (i) the prior model $P_2(\cdot)$ and (ii) the data.

Design a **non-iterative** / **direct** / **closed-form** implementable algorithm to solve the optimization problem. For instance, gradient-descent optimization and simulated annealing are **iterative** optimization methods producing a sequence of improving approximate solutions where sequence's length depends on the data values (*not* just data size or dimension). In contrast, algorithms for standard matrix decompositions or matrix inverses (when they exist) are **direct** methods that require a finite sequence of operations independent of the data values (even though they may depend on data size or dimension).

